

By defn. of universal quantifier, $\forall x \exists y (P(x) \wedge Q(y))$ is True.

Let, $\forall x \exists y (P(x) \wedge Q(y))$ be True.

By defn. of universal quantifier, for all elements a in the domain, $\exists y (P(a) \wedge Q(y))$ is True. By defn. of existential quantifier, there exists an element b in the domain for all elements a in the domain s.t.

$P(a) \wedge Q(b)$ is True. By defn. of AND, $P(a)$ is True for all elements a in the domain and $Q(b)$ is True for some element b in the domain.

By defn. of universal quantifier, $\forall x P(x)$ is True. By defn. of existential quantifier, $\exists x Q(x)$ is True. By defn. of AND, $\boxed{\forall x P(x) \wedge \exists x Q(x)}$ is True.

$$\therefore \forall x P(x) \wedge \exists x Q(x) \equiv \forall x \exists y (P(x) \wedge Q(y))$$

b) Show that $\forall x P(x) \vee \exists x Q(x)$ is equivalent to $\forall x \exists y (P(x) \vee Q(y))$, where all quantifiers have the same nonempty domain.

Let, $\forall x P(x) \vee \exists x Q(x)$ be False. By defn. of OR, $\forall x P(x)$ is False and $\exists x Q(x)$ is False. By defn. of universal quantifier, there exists some element a in the domain such that $P(a)$ is False. By defn. of existential quantifier, for all elements b in the domain, $Q(b)$ is False. By defn. of OR, for some element a in the domain and for all elements b in the domain, $(P(a) \vee Q(b))$ is False. By defn. of existential quantifier, $\exists y (P(a) \vee Q(y))$ is False. By defn. of universal quantifier, $\forall x \exists y (P(x) \vee Q(y))$ is False.

Let, $\forall x \exists y (P(x) \vee Q(y))$ be False. By defn. of universal quantifier, for some element a in the domain, $\exists y (P(a) \vee Q(y))$ is False. By defn. of existential quantifier, for all elements b in the domain, $(P(a) \vee Q(b))$ is False. By defn. of OR, $P(a)$ is False for some element a in the domain and $Q(b)$ is False for all elements b in the domain. By defn. of universal quantifier, $\forall x P(x)$ is False. By defn. of existential quantifier, $\exists x Q(x)$ is False. By defn. of OR, $\boxed{\forall x P(x) \vee \exists x Q(x)}$ is False.